

ES 1:

$$\begin{array}{c} \vdash A \rightarrow (B \rightarrow A) \quad [A]^1 \\ \frac{[A]^1}{B \rightarrow A} \rightarrow I^2 \\ \frac{B \rightarrow A}{A \rightarrow (B \rightarrow A)} \rightarrow I^1 \end{array}$$

ES 2:

$$\begin{array}{c} \vdash (A \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow (A \rightarrow C)) \quad [A \rightarrow (B \rightarrow C)]^1 \\ \frac{[A]^2 \quad [A \rightarrow (B \rightarrow C)]^2}{B \rightarrow C} \rightarrow E \quad [B]^2 \quad [A]^3 \\ \frac{B \rightarrow C}{C} \rightarrow I^3 \\ \frac{C}{A \rightarrow C} \rightarrow I^2 \\ \frac{A \rightarrow C}{B \rightarrow (A \rightarrow C)} \rightarrow I^1 \\ \frac{B \rightarrow (A \rightarrow C)}{(A \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow (A \rightarrow C))} \rightarrow I^1 \end{array}$$

ES 3:

$$\begin{array}{c} \vdash \neg(A \wedge \neg A) \quad [A \wedge \neg A]^1 \\ \frac{[A \wedge \neg A]^1}{A} \wedge E_1 \quad \frac{[A \wedge \neg A]^1}{\neg A} \wedge E_2 \\ \frac{A \quad \neg A}{\perp} \neg E \\ \frac{\perp}{\neg(A \wedge \neg A)} RAA^1 \end{array}$$

ES 5:

$$\begin{array}{c} \vdash (A \vee B) \rightarrow (B \vee A) \quad [A \vee B]^1 \\ \frac{[A]^2}{B \vee A} \vee I_1 \quad \frac{[B]^3}{B \vee A} \vee I_2 \\ \frac{B \vee A}{(A \vee B) \rightarrow (B \vee A)} \rightarrow I^1 \quad \vee E^{2,3} \end{array}$$

ES 6:

$$\begin{array}{c} \vdash (\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B) \quad [\neg B \rightarrow \neg A]^1 \\ \frac{[A]^2 \quad [\neg B \rightarrow \neg A]^1}{\neg B} \rightarrow E \\ \frac{\neg B}{A} \neg E \\ \frac{A}{\neg A} \neg E \\ \frac{\perp}{B} RAA^u \\ \frac{B}{A \rightarrow B} \rightarrow I^2 \\ \frac{A \rightarrow B}{(\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)} \rightarrow I^1 \end{array}$$

ES 7:

$$\vdash ((A \rightarrow B) \rightarrow A) \rightarrow A$$

$$\begin{array}{c}
 \frac{[\neg A]^2 \quad [A]^3}{\neg E} \\
 \frac{\frac{\perp}{B} \quad \perp E}{A \rightarrow B} \rightarrow I^3 \\
 \frac{[(A \rightarrow B) \rightarrow A]^1 \quad A \rightarrow B}{A} \rightarrow E \\
 \frac{A \quad [\neg A]^2}{\neg E} \\
 \frac{\frac{\perp}{A} \text{RAA}^2}{\neg E} \\
 \frac{}{((A \rightarrow B) \rightarrow A) \rightarrow A} \rightarrow I^1
 \end{array}$$

$$[(A \rightarrow B) \rightarrow A]^1$$

$$[\neg A]^2 \quad [\neg B]^4$$

$$[A]^3$$

ES 8:

$$\vdash \neg(A \wedge B) \rightarrow (\neg A \vee \neg B)$$

$$\begin{array}{c}
 \frac{[\neg A]^3}{\neg A \vee \neg B} \vee I \\
 \frac{}{[\neg(\neg A \vee \neg B)]^2} \neg E \\
 \frac{\perp}{\frac{\perp}{A} \text{RAA}^3} \\
 \frac{A \wedge B}{\neg A \vee \neg B} \neg E \\
 \frac{}{\neg(A \wedge B) \rightarrow (\neg A \vee \neg B)} \rightarrow I^1
 \end{array}$$

$$[\neg(A \wedge B)]^1$$

$$[\neg(\neg A \vee \neg B)]^2$$

$$[\neg B]^4$$

$$\neg A \vee \neg B$$

$$[\neg(\neg A \vee \neg B)]^2$$

$\neg E$

$$\perp$$

$$\frac{\perp}{B} \text{RAA}^4$$

$$\wedge I$$

$$[\neg(A \wedge B)]^1$$

$\neg E$

ES 9:

$$\vdash \neg(A \rightarrow B) \rightarrow (A \wedge \neg B)$$

$$\begin{array}{c}
 \frac{[A]^2 \quad [\neg A]^3}{\neg E} \\
 \frac{\perp}{B} \perp E \\
 \frac{B}{A \rightarrow B} \rightarrow I^2 \\
 \frac{A \rightarrow B \quad \neg(A \rightarrow B)}{\neg E} \\
 \frac{\perp}{A} \text{RAA}^3 \\
 \frac{A \wedge \neg B}{\neg(A \rightarrow B) \rightarrow (A \wedge \neg B)} \rightarrow I^1
 \end{array}$$

$$[B]^4$$

$$A \rightarrow B$$

$$[\neg(A \rightarrow B)]^1$$

$\neg E$

$$\frac{\perp}{\neg B} \text{RAA}^4$$

$$\wedge I$$

$$\vdash (A \rightarrow (B \vee C)) \rightarrow ((A \rightarrow B) \vee (A \rightarrow C))$$

ES 11:

$$\forall x (A(x) \rightarrow B(x))$$

ES 11:

ES 12:

$$\frac{[A(u) \wedge B(u)]^1}{\wedge E}$$

$$\frac{\exists x (A(x) \wedge B(x))}{\exists x A(x)} \quad \frac{A(u)}{\exists x A(x)} \exists I \quad \exists E$$

ES 13:

$$\vdash \forall x A(x) \rightarrow \exists x A(x)$$

$$\frac{\frac{\frac{[\forall x A(x)]^1}{\forall E} \quad A(t)}{\exists I} \quad \exists x A(x)}{\rightarrow I^1} \quad \forall x A(x) \rightarrow \exists x A(x)$$

$$[\forall x A(x)]^1$$

ES 14:

$$\vdash \forall x \neg A(x) \rightarrow \neg \exists x A(x)$$

$$\frac{\frac{[\forall x \neg A(x)]^1}{\forall E} \quad \neg A(u)}{\perp} \quad \frac{[A(u)]^3}{\neg E} \quad \frac{\perp}{\exists I^3} \quad \frac{\perp}{\neg \exists x A(x)} \quad \frac{\neg \exists x A(x)}{\rightarrow I^1} \quad \forall x \neg A(x) \rightarrow \neg \exists x A(x)$$

$$[\forall x \neg A(x)]^1$$

$$[\exists x A(x)]^2$$

ES 15:

$$\vdash \neg \exists x A(x) \rightarrow \forall x \neg A(x)$$

$$\frac{\frac{[\neg \exists x A(x)]^1}{\neg E} \quad \frac{[A(x)]^2}{\exists I} \quad \frac{\perp}{\neg \exists x A(x)} \quad \frac{\perp}{\forall I} \quad \frac{\forall x \neg A(x)}{\rightarrow I^1} \quad \neg \exists x A(x) \rightarrow \forall x \neg A(x)$$

$$[\neg \exists x A(x)]^1$$

$$[A(x)]^2$$

ES 16:

$$\begin{array}{c} \forall x (A(x) \rightarrow B(x)), \forall x (B(x) \rightarrow C(x)) \vdash \forall x (A(x) \rightarrow C(x)) \\ \forall x (A(x) \rightarrow B(x)) \quad \forall E \\ \hline A(x) \rightarrow B(x) \quad [A(x)]^1 \rightarrow E \quad \forall x (B(x) \rightarrow C(x)) \quad \forall E \\ \hline B(x) \quad B(x) \rightarrow C(x) \quad \rightarrow E \\ \hline C(x) \quad \rightarrow I^1 \\ \hline A(x) \rightarrow C(x) \quad \forall I \\ \hline \forall x (A(x) \rightarrow C(x)) \end{array}$$

ES 17:

$$\begin{array}{c} \vdash \exists x \neg A(x) \rightarrow \neg \forall x A(x) \\ [\forall x A(x)]^2 \quad \forall E \\ \hline A(u) \quad [\neg A(u)]^3 \quad \neg E \\ \hline \perp \\ \hline [\exists x \neg A(x)]^1 \quad \exists E^3 \\ \hline \perp \quad RAA^2 \\ \hline \neg \forall x A(x) \quad \rightarrow I^1 \\ \hline \exists x \neg A(x) \rightarrow \neg \forall x A(x) \end{array}$$

$[\exists x \neg A(x)]^1$
 $[\forall x A(x)]^2$
 $[\neg A(u)]^3$

ES 18:

$$\begin{array}{c} \forall x (A(x) \rightarrow \neg B(x)) \vdash \neg \exists x (A(x) \wedge B(x)) \\ [A(u) \wedge B(u)]^2 \quad \wedge E \quad \forall x (A(x) \rightarrow \neg B(x)) \quad \forall E \\ \hline A(u) \quad A(u) \rightarrow \neg B(u) \quad \rightarrow E \\ \hline \neg B(u) \\ \hline \perp \quad \neg E \\ \hline [\exists x (A(x) \wedge B(x))]^1 \quad \exists E^2 \\ \hline \perp \quad RAA^1 \\ \hline \neg \exists x (A(x) \wedge B(x)) \end{array}$$